Programming(2)

Sayed Mostafa Modaghegh



fall semester 2025 _ Gharjistan uni

Learning Objectives

- ✓ Compare values using comparison operators
- ✓ Combine conditions with logical operators
- ✓ Use if and else for decision making
- ✓ Understand type conversion (casting)
- ✓ Write simple decision-based programs



Casting

Casting



Use casting when you need to convert data types.

index.py > ...

```python
# for casting:
# new variable = datatype(old variable)

x = int("5")
y = float(2)
z = str(10)

print(x + y)      # 7.0
print("Number:", z) # Number: 10

```

Comparison Operators



Comparison Operators

always return True or False

Comparison Operators

Operator	Meaning
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

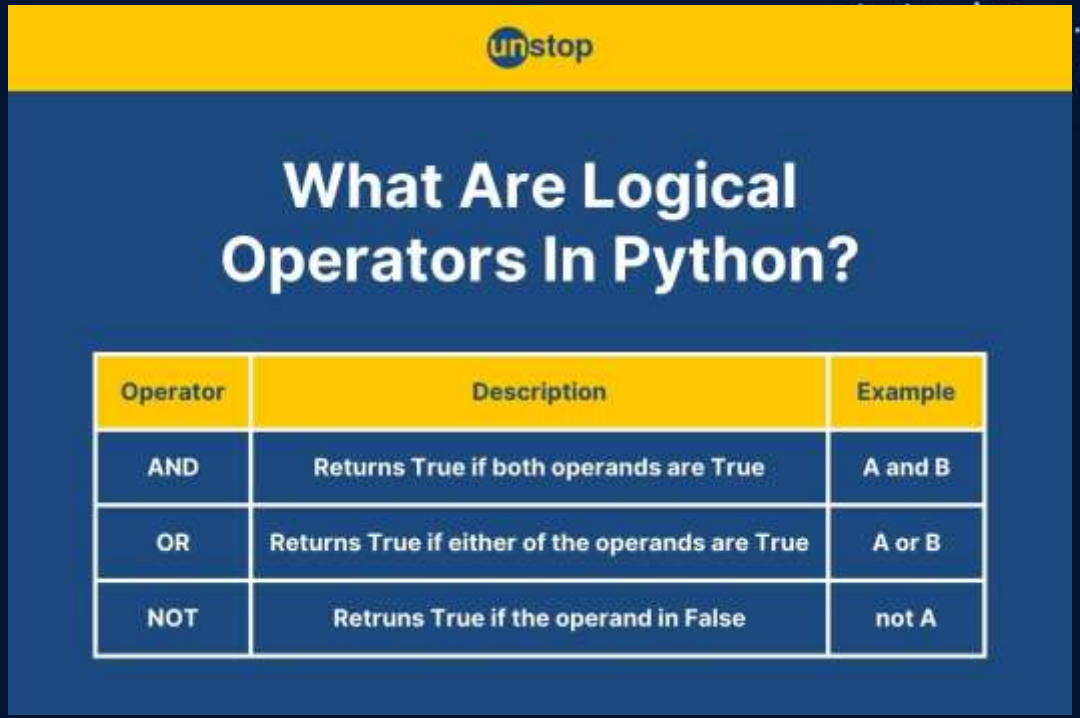


Logical Operators



Logical Operators

always return True or False





If statement



If statement

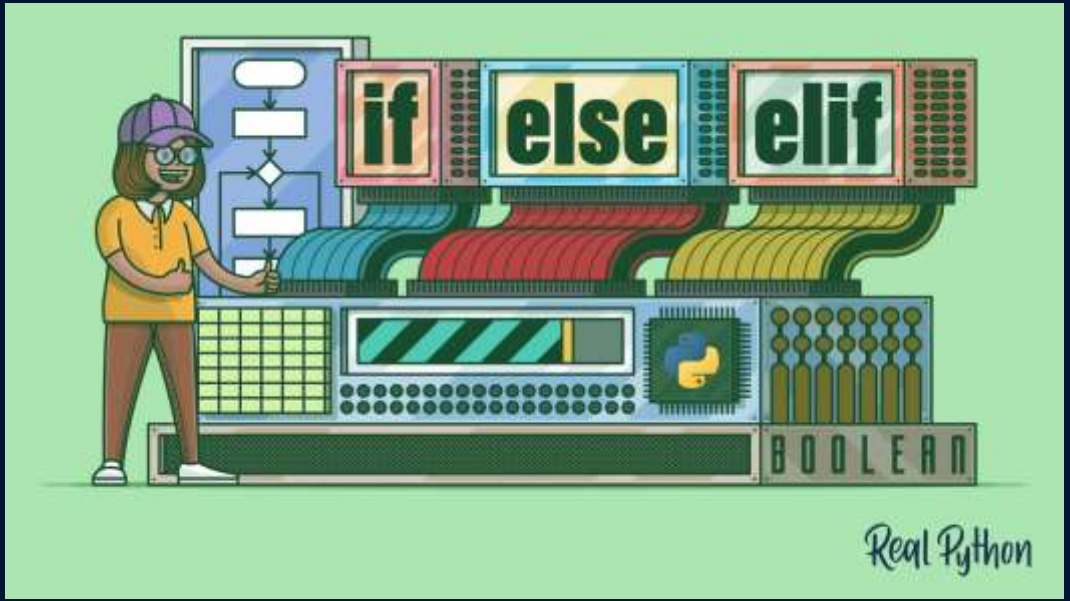
What is an *if statement*?

The if statement is used :

when we want the program to make a decision.

It checks a condition, and if that condition is True, a block of code runs.

If it's False, that block is skipped (or another block runs using else).



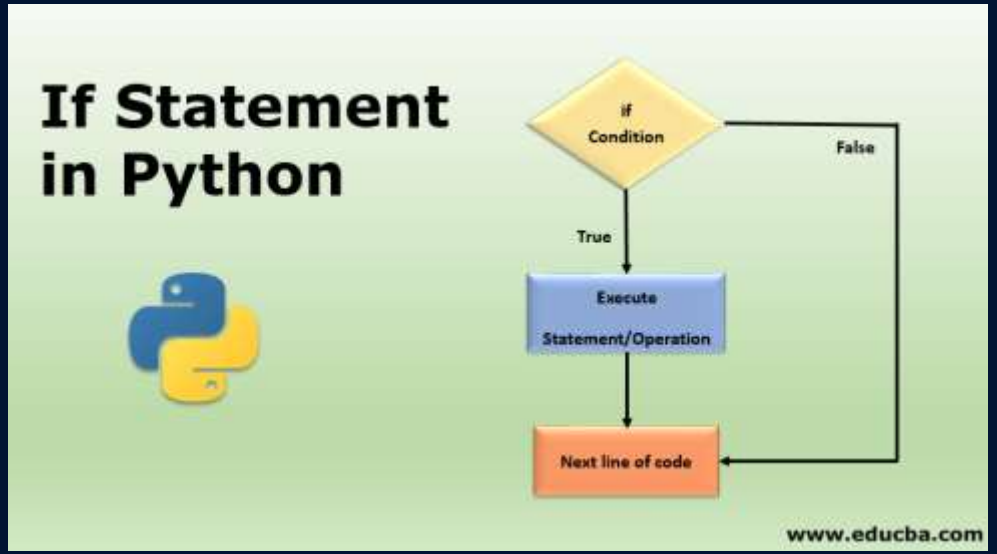
If statement

Tips for Writing Conditions

Use indentation (4 spaces) after if and else

Always end each condition line with :

Python cares about spacing!



If statement

If example

index.py > ...

```python
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

if a > b:
    print("First number is greater")
else:
    print("Second number is greater or equal")

```



If statement

Combining Conditions

index.py > ...

```python
score = int(input("Enter your score: "))

if score >= 90:
    print("Excellent!")
elif score >= 60 and score < 90:
    print("Passed.")
else:
    print("Failed.")

```



Thank You

About me:
Sayed Mostafa Modaghegh
AI and Robotics student
Computer vision developer



Sm.mostafa79modaqeq@gmail.com

@mostafa_modaghegh